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Q1: Library Management (Mini System)

Code

```cpp
#include<iostream>
using namespace std;
int totalIssued(int books[3][3]) {
    int issued=0;;
    for(int i=0;i<3;i++){
        issued=issued+books[i][2];
        }
    return issued;
    }
void issueBook(int books[3][3], int id){
        for(int i=0;i<3;i++){
            if(id==books[i][0]){
                if(books[i][1]>0){
                    books[i][2]=books[i][2]+1;
                    books[i][1]=books[i][1]-1;
                }
                else{
                    cout<<"Not Enough"<<endl;
                }
            }
        }
    }
void printArr(int books[3][3]){
    cout<<"ID";
    cout<<"\t     Available";
    cout<<"\t     Issued"<<endl;
    for(int i=0;i<3;i++){
    for(int j=0; j<3;j++){
            cout<<books[i][j]<<"\t\t";
            }
        cout<<endl;
        }
    }
```

```cpp
int main(){
    int books[3][3]={
    {2445,10,0},
    {2446,10,0},
    {2447,10,0}
    };
    int fines[3] = {0, 0, 0};
    issueBook(books, 2445);
    printArr(books);
    int total=totalIssued(books);
    cout<<"Total Books Issued: "<<total;
}
```

Q1: Library Management (Mini System)

Output

```
ID           Available       Issued
2445            9               1
2446            10              0
2447            10              0
Total Books Issued: 1
--------------------------------
Process exited after 0.1538 seconds with return value 0
Press any key to continue . . .
```

Q2: Temperature Data Processing

Code

```cpp
#include<iostream>
using namespace std;
double average(int arr[2][7]){
    int sum=0;
    for(int i=0; i<2;i++){
        for(int j=0; j<7;j++){
            sum=sum+arr[i][j];
        }
    }
    double ave=sum/14;
    return ave;
}
int main(){
    int tempWeek[7];
    int detailedTemp[2][7];
    int cold=0;
    int hot=0;
    for(int i=0; i<7;i++){
        cout<<"Enter temprature for day"<<i+1<<endl;
        cin>>tempWeek[i];
    }
    for(int i=0; i<7;i++){
        if(tempWeek[i]<15){
            cold=cold+1;
        } else if(tempWeek[i]>30){
            hot=hot+1;
        }
    }
    for(int i=0; i<2;i++){
        if(i==0){
            cout<<"Enter the Morning tempratures"<<endl;
        } else{
            cout<<"Enter the Evening tempratures"<<endl;
        }
        for(int j=0; j<7;j++){
            cout<<"Enter temprature for day"<<j+1<<endl;
            cin>>detailedTemp[i][j];
        }
    }
    int hottest=tempWeek[0];
    int coldest=tempWeek[0];
    double aveTemp=average(detailedTemp);
    for(int i=0; i<7;i++){
        if(tempWeek[i]>hottest){
            hottest=tempWeek[i];
        }
    }
    for(int i=0; i<7;i++){
        if(tempWeek[i]<coldest){
            coldest=tempWeek[i];
        }
    }
    cout<<"Hottest Day: "<<hottest<<endl;
    cout<<"Coldest Day: "<<coldest<<endl;
    cout<<"Average :"<<aveTemp;
}
```

Q2: Temperature Data Processing

Output

```
Enter temprature for day 1
24
Enter temprature for day 2
25
Enter temprature for day 3
43
Enter temprature for day 4
25
Enter temprature for day 5
26
Enter temprature for day 6
42
Enter temprature for day 7
37
Enter the Morning tempratures
Enter temprature for day 1
14
Enter temprature for day 2
16
Enter temprature for day 3
12
Enter temprature for day 4
17
Enter temprature for day 5
14
Enter temprature for day 6
19
Enter temprature for day 7
13
Enter the Evening tempratures
Enter temprature for day 1
24
Enter temprature for day 2
27
Enter temprature for day 3
29
Enter temprature for day 4
30
Enter temprature for day 5
33
Enter temprature for day 6
29
Enter temprature for day 7
24
Hottest Day: 43
Coldest Day: 24
Average :21
--------------------------------
```